

Cost-Shifting and Competition^{*}

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Abstract

Hospital cost shifting—recouping public payment cuts by charging private payers more—is a claim basic economics rejects. We show why, and when the rejection fails. With the set of hospitals fixed, a public-payment cut moves neither the surplus a hospital and insurer bargain over nor the power dividing it, so commercial prices do not move, whatever the insurer’s market power. Cost-shifting requires that the cut threaten a hospital’s survival: the insurer then either keeps it open at a higher price, or lets it close while its surviving rivals raise theirs. Pass-through appears only near exit and is zero for a secure hospital’s own price. Empirically, the commercial-price response does not scale with insurer market power, but rises with a hospital’s predicted closure risk, and closures raise rivals’ prices. The exit margin can reconcile a mixed literature.

Keywords: cost-shifting, hospital pricing, insurer bargaining power, provider financial distress, endogenous market structure

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1 Introduction

Hospitals have long argued that when Medicare or Medicaid cuts reimbursement, they must recover the shortfall by charging private insurers more. This claim, often referred to as “cost-shifting”, anchors hospital lobbying, insurer complaints, and a large policy literature.¹ However, most economists regard cost-shifting as theoretically unfounded. A profit-maximizing hospital already sets private prices to maximize private profit; a cut to public payments does not change the optimal price, so the hospital cannot shift their losses elsewhere.

Even in the more modern models, where hospitals and insurers bargain over the total surplus from the market, the available surplus is determined by the insurer-consumer market structure and the hospitals’ marginal cost for privately insured patients. The hospitals’ total cost does not enter the equation. Thus far, economists have had to turn to “non-traditional” assumptions, such as utility-maximizing nonprofits that cross-subsidize (Hay, 1983), price discrimination across payers (Showalter, 1997), or free-riding on other insurers (Glazer and McGuire, 2002), for why public costs occasionally seem to influence private prices.²

In this paper, we first show theoretically that a very traditional economic assumption – $\text{profit} \geq 0$ – naturally induces cost shifting in some markets but not others. The intuition is simple: when the cut to public funds is small enough (or the hospitals already profitable enough) that the profit assumption is not binding, then the economist’s intuition passes through unscathed and no cost shifting occurs. However, when a cut makes at least one hospital unviable under the current negotiated terms, insurers face a choice. They can keep that hospital open by agreeing to a higher reimbursement, or they can let that hospital close and re-negotiate with the surviving hospitals, who now have more market power and can thus command higher prices. In either case, cost-shifting will be observed.

¹For a recurring example, the American Hospital Association’s “Underpayment by Medicare and Medicaid” fact sheet reports that Medicare and Medicaid pay hospitals below the cost of care—82 cents per dollar of Medicare cost in 2022—the premise of the industry’s argument that public shortfalls are recouped from other payers. See <https://www.aha.org/fact-sheets/2020-01-07-fact-sheet-underpayment-medicare-and-medicaid>.

²The empirical record is genuinely mixed: some studies find substantial pass-through (Wu, 2010), others find none or the opposite sign (Wagner, 2016; Dranove, 1988), and reviews conclude only that the magnitude depends on the market (Frakt, 2011; Morrissey, 2003).

We then test our theory empirically using 2018 340B outpatient cut, a Medicare rule that sharply reduced reimbursement for the outpatient drugs of participating hospitals. Three findings line up with the theory. First, where the participation constraint is slack, the cut does not reach commercial prices at all, and it does not reach them any more where insurers or hospitals hold more market power: the price response is flat in concentration, at 0.0088 log points per standard deviation. Second, a hospital passes the cut into its own price only when it is close to failing. A one-standard-deviation increase in predicted closure risk makes the own-price response 0.0261 log points more positive, hospitals far from the exit margin show -0.0271 log points, and pooling the two groups gives a null. Third, when exit is realized rather than averted, the prices that move are those of a different, safer hospital: commercial prices at survivors rise by 3.9% after a rival closes in the same metropolitan market, and by more when the closing hospital took more of the market with it, even though nothing has happened to the survivors' own public revenue. Inpatient prices, which the outpatient cut does not touch, do not move.

The debate over cost-shifting is old and has swung more than once. It began as an accounting claim: hospitals reported that public payers reimbursed below cost and that the gap was recovered from private payers, an argument that carried weight in the 1980s, when Medicare moved to prospective payment and several states set hospital rates for all payers at once. Economists objected that a hospital maximizing private profit has no such lever (Dranove, 1988; Morrissey, 1994), and the empirical work that followed never settled the question. Studies of the Balanced Budget Act cuts found substantial pass-through (Wu, 2010), studies of insurance-mix shocks found none (Wagner, 2016), and studies of Medicare rate variation found private prices moving in the same direction as public ones (White, 2013; Clemens and Gottlieb, 2017). Reviews concluded that cost-shifting was smaller and more contingent than the industry claimed, without saying what it was contingent on (Morrissey, 2003; Frakt, 2011). The most direct recent evidence finds pass-through at hospitals with market power and none on average (Dranove et al., 2017). What the debate has never had is a condition that says which hospitals should shift and which should not.

Before going to the exit margin, we check the part of the theory that says cost-shifting should not be there at all. The folk version of cost-shifting is a story about bargaining power, so if it were right, a public cut would raise commercial prices most where insurers are most concentrated. It does not. Interacting the 340B cut with commercial insurer concentration leaves the price response flat, at 0.0088 log points per standard deviation

of concentration, and the estimate is tight enough to rule out an insurer-power channel any larger than the exit-margin effect we report next. We get the same answer measuring power by hospital concentration and by the gap between the two. Market power decides who keeps the surplus, not whether a public cut ever enters it.

The main test uses the 2018 340B outpatient cut, a Medicare rule that dropped the payment for 340B-acquired outpatient drugs from average sales price plus 6 percent to average sales price minus 22.5 percent (Centers for Medicare and Medicaid Services, 2017). It is a large and abrupt cut to the public revenue of one identifiable set of hospitals, and it lets us compare hospitals that were close to failing when it landed against otherwise similar hospitals that were comfortable. We measure how close a hospital was to failing with a closure model fit on closures that predate the cut, using financial information older still, so the measure cannot be a product of the shock it is used to study. A one-standard-deviation increase in predicted closure risk makes the commercial outpatient price response to the cut 0.0261 log points more positive. At the hospitals with low predicted closure risk the response is -0.0271 log points, so where the theory says there should be no cost-shifting, there is none. Pooling the two groups together gives a null, which is what a study that ignored the exit margin would have reported. Inpatient prices, which the outpatient cut does not touch, do not move.

The other branch of the theory covers the hospitals that do not survive. When a hospital closes, the insurer has one fewer hospital to contract with and the ones that remain are harder to replace, so their negotiated prices should rise even though nothing has happened to their own public revenue. Matching each surviving hospital to the closures in its market, we find that commercial prices at the survivors rise by 3.9% after a closure in the metropolitan market and by 5.3% using the tighter county definition, and that the increase is larger the more of the market the closing hospital took with it. Markets that are about to lose a hospital are not already priced above markets that never lose one, so the increase follows the closure rather than anticipating it. The hospital that closes recovers nothing. The gain goes entirely to the hospitals left standing.

The paper contributes to three literatures. First, we give the cost-shifting literature a condition that can reconcile its conflicting estimates. Decades of empirical work have disagreed over whether hospitals raise private prices when public payments fall, reaching opposite conclusions from broadly similar data (Dranove, 1988; Morrissey, 2003; Wu,

2010).³ Another strand finds the opposite sign entirely, with lower public rates followed by lower private rates for hospitals (White, 2013) and physicians (Clemens and Gottlieb, 2017). If that co-movement came from benchmarking or reference pricing it would show up uniformly across providers, while our mechanism puts it only at the hospitals near exit, so the two are distinguishable. On the theory side, cost-shifting has been rationalized only by giving up private profit maximization, through nonprofit objectives, scope economies, or free-riding across payers (Hay, 1983; Showalter, 1997; Glazer and McGuire, 2002).⁴ We keep profit maximization and put cost-shifting at the entry-exit margin instead. Pass-through then appears where hospitals are near exit and vanishes where they are secure, so two studies drawing samples of different financial composition can reach opposite conclusions without either being wrong. It also fits the finding that the 2008 stock-market collapse produced cost-shifting only at hospitals with substantial market power and none on average (Dranove et al., 2017), since a large negative wealth shock is exactly what pushes hospitals up against the participation constraint where our mechanism says pass-through starts.

Second, we make a conceptual point for the literature on hospital-insurer bargaining. That literature treats each hospital-insurer pair as a separate negotiation, and treats administered public prices as sitting outside those negotiations entirely. Entry and exit undo both separations. Whether a hospital stays open is itself an outcome of the bargain, so a cut to one hospital's public revenue can move the price a different hospital negotiates with a different insurer, and a public payment policy can raise commercial prices without ever entering the bargaining problem directly. Where participation is not at stake, neither spillover exists and public payments wash out of the negotiated price, whatever the insurer's power (Proposition 1). This links cost-shifting to the bilateral-bargaining framework that now organizes the analysis of hospital and insurer competition (Gowrisankaran et al., 2015; Collard-Wexler et al., 2019; Gaynor and Vogt, 2000)⁵, and to evidence that provider exit and changes in market structure move negotiated prices (Carroll and Chang, 2025; Lindrooth et al., 2018; He and Mellor, 2012).

Third, we speak to the literature on what drives the cost of commercial insurance. Hospital care is the largest single component of what a commercial plan pays out, and the prices behind it vary enormously across hospitals treating similar patients (Reinhardt,

³See also Morrissey (1994), Frakt (2011), Wagner (2016), and Melnick and Fonkych (2008).

⁴See also Santerre (2005).

⁵See also Gaynor and Vogt (2003), Gaynor and Town (2011), and Dafny (2010).

2006; Whaley et al., 2024). Almost all of the explanation for that variation has been looked for in market structure, on both sides of the bargain: hospital consolidation raises negotiated prices (Gowrisankaran et al., 2015; Collard-Wexler et al., 2019), hospital closures raise the prices of the hospitals that remain (Carroll and Chang, 2025; Lindrooth et al., 2018), and insurer concentration moves what plans charge (Dafny, 2010). What plans charge is in turn borne by workers, through wages and through which coverage their employer offers at all (Gao et al., 2022; Wu, 2025). We add a determinant that is not a feature of market structure: the financial condition of the particular hospitals in an insurer’s network. A hospital close enough to failing that its participation is genuinely at stake extracts a higher price, and if it fails anyway the hospitals that absorb its patients extract more still. That connects the hospital–insurer bargain to a broader literature on how a firm’s own financial condition shapes the prices it charges and the services it delivers (Ge, 2022; Erel et al., 2025). It also says something specific about where the cost of a public-payment cut lands: not on commercial premiums generally, but on the plans whose networks contain fragile hospitals, and in proportion to how fragile those hospitals are.

The distinction matters directly for policy. If hospitals shift costs, then cuts to public reimbursement raise commercial prices as an unintended byproduct, and the public and private payment systems cannot be set in isolation from each other. If they do not, the two systems can be governed separately. Our answer is that both are true, for different hospitals. Payment changes spill into commercial prices where hospitals are financially fragile and leave them alone where hospitals are secure, so which regime a market is in depends on how close its hospitals sit to the exit margin. That margin is not fixed either: it is moved by the same public payment policy whose spillovers are at issue.

2 Theory

This section makes one argument: in a profit-maximizing model, a cut to public reimbursement reaches commercial prices only through the entry and exit of hospitals. We start by fixing a measure of cost-shifting and a model of how commercial prices get set, with insurers competing for enrollees and hospitals and insurers bargaining over reimbursement. We then prove a neutrality result. Hold the set of hospitals fixed, and a change in public reimbursement moves neither the surplus the parties bargain over nor their bargaining power, so commercial prices do not move either, no matter how market

power is distributed. Cost-shifting requires that the shock threaten a hospital's participation. Once it does, the insurer has to choose. It can keep the hospital open on richer terms, or it can let the hospital go and deal with the ones that remain. Either way a commercial price rises. Those two branches, and the neutrality that holds away from them, organize the empirical work that follows.

2.1 Baseline Model and Cost-Shifting Measure

Following Hay (1983) and Showalter (1997), we model hospitals as two-output firms serving distinct markets. In the private market, prices and quantities are determined through economic equilibrium. In the public market, the government sets prices by fiat, and hospitals must meet all demand at these regulated prices.

2.1.1 Model Setup

Consider a hospital with the following profit function:

$$\pi = q_m (p_m - c_m) + q_h (p_h - c_h) - F \quad (1)$$

where subscript m denotes the public (e.g., Medicare) market and subscript h denotes the private (e.g., commercial insurance) market. Variables q_i , p_i , and c_i represent quantities, prices, and marginal costs in market $i \in \{m, h\}$, respectively, and F represents fixed costs.

We initially assume:

1. Hospitals maximize profits
2. Constant returns to scale in variable costs
3. No economies of scope between public and private production
4. Public prices p_m are exogenously set by government

2.1.2 Defining Cost-Shifting

Cost-shifting refers to the equilibrium response of private prices to changes in public reimbursement rates. We define our primary measure of cost-shifting as:

$$\theta = -\frac{\partial p_h}{\partial p_m} \frac{p_m}{p_h} \frac{p_h q_h}{p_m q_m}$$

This measure captures the elasticity of private price with respect to public price, weighted by the ratio of private to public revenue. When $\theta = 1$, we have full cost-shifting: a reduction in public revenue is completely offset by increased private revenue (holding quantities constant). When $\theta = 0$, there is no cost-shifting.⁶

2.2 Surplus, Bargaining, and the No-Cost-Shifting Benchmark

The intuition that cost-shifting is theoretically dubious can be seen quickly and then stated sharply. The quick version is the textbook one. Suppose hospitals compete à la Bertrand in the private market and that both q_m and p_m are set by fiat. Maximizing the hospital's profits in equation 1 in p_h leads to the standard pricing equation:

$$p_h = c_h - \frac{q_h}{\partial q_h / \partial p_h} \quad (2)$$

If the hospital is a monopolist, then this equation serves to define the equilibrium private price. If there are multiple (presumably differentiated) hospitals operating in the market, then the first order conditions for those hospitals (expressed by this same equation, one for each hospital) generate a system of equations sufficient (under some regularity conditions on demand) to generate an equilibrium set of prices.

⁶Alternative measures are possible under different assumptions about quantity responses. For instance, if we allow private quantity to adjust while holding public quantity fixed:

$$\hat{\theta} = -\frac{\partial p_h}{\partial p_m} \frac{p_m}{p_h} \frac{p_h q_h}{p_m q_m} (1 + \epsilon^d)$$

where ϵ^d is the elasticity of private quantity with respect to private price. Throughout this analysis, we use θ as our primary measure, and all derivatives should be interpreted as equilibrium responses.

Either way, none of its terms is a function of p_m : any private price p_h that solves this equation at one public price p_m also solves it at any other public price p_m^* , so p_m cannot affect p_h . Moving to perfect competition would make things no better, since the second term on the right-hand side of equation 2 would then be eliminated.

The quick version invites a natural objection: hospital prices are not set unilaterally but negotiated with insurers, and negotiation is exactly where the folk theory of cost-shifting locates the action—the hospital points to its public shortfall as a reason for a higher private rate. Yet the benchmark survives negotiation, and even insurer market power, once the set of hospitals is held fixed. Model the commercial price as the outcome of a hospital–insurer bargain that splits the surplus their agreement creates, with the hospital’s public net revenue $R \equiv (p_m - c_m)q_m$ part of its payoff. We develop this bargaining model in Section 2.4.2; here we state what it delivers at the benchmark.

Proposition 1 (Fixed-Participation Neutrality). *Suppose the commercial price is set by Nash bargaining between a hospital and an insurer with hospital bargaining weight $\alpha \in [0, 1]$, and the set of operating hospitals is fixed, so that a hospital which rejects the contract keeps operating and retains its public net revenue R . Then $\partial p_h / \partial p_m = 0$ for every α : there is no cost-shifting.*

A proof is given in Appendix A. When a hospital would survive a breakdown in talks (rejecting the contract but keeping its public business), its public revenue R enters its agreement and disagreement payoffs identically and cancels from the surplus the parties divide, so the negotiated price does not move with p_m , whatever the hospital’s bargaining power. A public cut could in principle reach the negotiated price in one of two ways, by changing the surplus the parties divide or by changing how they divide it. With the set of hospitals fixed it does neither. Market power does not rescue cost-shifting here. It only decides who keeps the surplus, not whether public reimbursement ever enters it. We prove neutrality for a monopoly insurer, taking its value of an agreement to be the full enrollee surplus. The result extends a fortiori to richer market structures: what drives it is that a surviving hospital’s disagreement payoff is invariant to p_m , since public revenue R enters its agreement and disagreement payoffs identically whatever the number of hospitals or insurers, so competition changes the surplus and its division but not whether p_m enters the bargain. This is our benchmark, and it is the sense in which cost-shifting is not a general feature of profit-maximizing pricing.

2.3 Existing Escapes from the Neutrality Result

The literature escapes this benchmark only by leaving one of its premises—in each case, by relaxing profit maximization or its cost structure. One way out of the no-cost-shifting result is to introduce economies of scope between public and private output and to relax the assumption that hospitals must treat all comers in the public program. The models of Hay (1983) and Showalter (1997) is of this type. Examining again the pricing equation above, consider that a reduction in p_m causes the hospital to reduce q_m . If there are positive scope economies, then this will increase c_h , leading to higher private prices, i.e. there will be cost-shifting. Of course if there are diseconomies of scope, then there will be anti-cost-shifting. Because of the way Hay (1983) and Showalter (1997) specify costs, there can be economies of scope only if the hospital is operating in the increasing-returns portion of its cost-function.⁷ This is possible only for hospitals with market power, and the degree of scope economies is constrained by the degree of market power. Hay (1983), Showalter (1997), and subsequent interpreters have found it implausible that hospitals are operating in the increasing returns portion of their cost functions, so this mechanism is an unpopular way to rationalize cost-shifting.

Another traditional way out of the no-cost-shifting result is to introduce non-profit-maximizing behavior by the hospital. Adapting the models and borrowing the results of Dranove (1988), Gaynor and Vogt (2003), Santerre (2005), and Lakdawalla and Philipson (2006), assume that hospitals maximize a utility function increasing in profit and in output, $U(\pi, q_m + q_h)$ subject to a constraint that they not lose too much money, $\pi \geq -\bar{\pi}$. Examine again the first-order condition for private pricing and denote by λ the lagrange multiplier on the not-too-big-losses constraint:

$$p_h = c_h - \frac{\frac{\partial U}{\partial Q}}{\frac{\partial U}{\partial \pi} + \lambda} - \frac{q_h}{\partial q_h / \partial p_h} \quad (3)$$

There is an extra term now in the pricing equation. This term represents the fact that, because not-for-profit hospitals value output per se, they act like for-profits with lower costs. This behavior is tempered by the constraint, should it be binding. Now, if the government lowers p_m , this causes a negative shock to profits. This may affect private prices

⁷This model assumes that costs are $C(q_m + q_h)$ which implies that scale and scope economies are the same thing, i.e. that there are positive economies of scale if and only if there are positive economies of scope.

in two ways. First, with a concave utility function, lower profits causes the profit derivative of U to rise, reducing the absolute value of the second term. Second, lower profits make it more likely that the profit constraint is binding, and, should it be binding to begin with, increases the value of λ . This also decreases the absolute value of the second term above. Both effects push towards higher private prices and, thus, to cost-shifting. These escapes work, but each buys cost-shifting by giving up profit maximization or constant returns; we now show that neither is needed once the set of hospitals is allowed to change.

2.4 Cost-Shifting at the Exit Margin

Fixed-participation neutrality has a troubling loose end. If the public payer lowers reimbursements and no cost-shifting occurs, then the hospital simply makes less profit than before, and nothing in the model above limits how far this can go. Can the public payer keep lowering reimbursements without limit, secure that nothing untoward happens beyond making hospital owners poorer? It cannot, and seeing why points to the only channel through which public cuts reach commercial prices. Push reimbursement down far enough and some hospital is pushed into losses and toward exit. At that point neutrality breaks, because the premise of Proposition 1, that the hospital survives disagreement, fails.

A cut that makes a hospital unviable on its current terms can resolve in two ways. The insurer can keep the hospital open on richer terms, so that the hospital's own price rises, which is an averted exit. Or it can let the hospital go and re-contract with the survivors, who are now scarcer and whose prices rise, which is a realized exit. Both are cost-shifting, and both need the exit margin to be live. Which one happens depends on whether keeping the hospital open is jointly worthwhile. We take the realized-exit branch first, through the market structure it changes, and then the averted-exit branch, through the bargain it shifts.

Which branch occurs is a comparison of surpluses. Keeping the distressed hospital open yields the ongoing-relationship total surplus $TS = W - c_h q_h + R - F$; letting it close lets the insurer re-contract with the scarcer survivors, whose improved outside options it must then pay for. The pair keeps the hospital open when ongoing-relationship surplus weakly exceeds the value of re-contracting, and lets it go otherwise. As a public cut lowers

R , ongoing-relationship surplus falls and the regime flips: a hospital the insurer would rather retain is pivotal and takes the averted-exit branch, while one it can cheaply replace is replaceable and takes the realized-exit branch.

2.4.1 Realized Exit: Surviving Rivals' Prices

If no agreement keeps the hospital open, it exits and the set of firms in the market falls by one. This is where neutrality's other premise, a fixed set of hospitals, gives way. With one fewer hospital to contract with, each survivor is harder for the insurer to replace, its outside option in bargaining improves, and its negotiated price rises. The hospital that exits recovers nothing; the commercial-price increase goes to the hospitals that remain. This is the long-run margin at which the public payer's power over prices runs out, and the natural way to analyze it is through the free-entry condition that ties market structure to prices.

Free Entry Equilibrium Normally, this kind of objection is overcome by endogenizing the number of firms in the market. If the public payer forces prices down far enough, we expect to see some firms pushed into losses and then into exit. Borrowing from Bresnahan and Reiss (1991), in the long run, market structure and prices are set by the free entry condition:

$$\Pi(N) = (p_m - c_m) \frac{Q_m}{N} + (p_h(N) - c_h) \frac{Q_h(p_h(N))}{N} - F(N) = 0$$

In this expression, $\Pi(N)$ is the profit which the N th entrant to the market makes, Q_m is the market level of demand for hospital services in the public market, $Q_h(p_h)$ is the market level of demand for hospital services in the private market, and $F(N)$ are the fixed costs paid by the N th entrant. The other notation is as it was previously.

The equality in the expression is the free-entry equilibrium condition, and this condition serves to implicitly define the equilibrium value of N , the number of firms in the market. Firms enter the market until the marginal one earns zero profit, assuming implicit-

itly that Π is decreasing in N . Considering that N is an integer, this equilibrium condition is not quite right, but we will ignore this problem and treat N as a continuous variable for the duration of this section.

In this model, the details of the pricing equilibrium after entry are left open. The equilibrium price in the private market, $P_h(N)$, is determined by whatever game the parties play—Bertrand, Cournot, or bargaining. Under bargaining, the dependence of P_h on N is precisely the survivor's-outside-option channel: fewer hospitals means each is harder for the insurer to replace, so $P_h(N)$ falls in N . The structure of the entry equation imposes a symmetric equilibrium in that game.

When a new firm enters the market, many of the terms determining Π change. First, the volume of services flowing to each firm falls, Q_m/N and Q_h/N decrease, as market demand is allocated over more firms. Second, in most models of oligopoly pricing market price, $p_h(N)$, is a decreasing function of the number of competitors. Third, this price decrease with entry induces an increase in market quantity demanded, $Q_h(p_h(N))$. Finally, one expects later entrants to pay larger fixed costs since earlier entrants have already claimed the most favorable local amenities (location, reputational first-mover advantage, etc).

In order to analyze cost-shifting in this context, perturb the equilibrium condition by lowering p_m , allow N to settle at a new, lower level and then analyze the effect of the new, lower N on market price, p_h .

To simplify the exposition, we rewrite the profit function using π_h to denote variable profits in the private market and π_m to denote variable profits in the public market:

$$\Pi(N) = \pi_m + \pi_h - F(N) = 0$$

To analyze the effects of changing p_m on equilibrium N , differentiate this object in p_m , keeping in mind that its terms vary in N and that p_m affects only the first term directly (i.e. not through its effect on N):

$$\frac{d\pi_m}{dp_m} + \left(\frac{d\pi_m}{dN} + \frac{d\pi_h}{dN} - \frac{dF}{dN} \right) \frac{dN}{dp_m} = 0$$

Solving for $\frac{dN}{dp_m}$ yields:

$$\frac{dN}{dp_m} = - \frac{\frac{d\pi_m}{dp_m}}{\frac{d\pi_m}{dN} + \frac{d\pi_h}{dN} - \frac{dF}{dN}}$$

Finally, notice that p_h is affected by N , so that:

$$\begin{aligned} \frac{dp_h}{dp_m} &= \frac{dp_h}{dN} \frac{dN}{dp_m} \\ &= - \frac{dp_h}{dN} \frac{\frac{d\pi_m}{dp_m}}{\frac{d\pi_m}{dN} + \frac{d\pi_h}{dN} - \frac{dF}{dN}} \end{aligned} \quad (4)$$

To calculate our measure of cost-shifting, we substitute the left-hand-side of this expression into the definition of θ and, after some algebra:

$$\begin{aligned} \theta &= - \frac{\partial p_h}{\partial p_m} \frac{p_m}{p_h} \frac{p_h q_h}{p_m q_m} \\ &= \left(\frac{dp_h}{dN} \frac{N}{p_h} \right) \cdot \frac{\left(\frac{d\pi_m}{dp_m} p_m \right)}{\left(\frac{d\pi_m}{dN} N \right) + \left(\frac{d\pi_h}{dN} N \right) - F \left(\frac{dF}{dN} \frac{N}{F} \right)} \cdot \frac{p_h q_h}{p_m q_m} \\ &= \epsilon_N^{p_h} \cdot \frac{p_m q_m}{-\pi_m + (-\pi_h + \epsilon_N^{p_h} (q_h p_h + \epsilon_M^d)) - F \epsilon_N^F} \cdot \frac{p_h q_h}{p_m q_m} \\ &= -\epsilon_N^{p_h} \cdot \frac{1}{\frac{\pi_m + \pi_h}{p_h q_h} - \epsilon_N^{p_h} + \epsilon_N^{p_h} \epsilon_M^d \frac{\pi_h}{p_h q_h} + \frac{F}{p_h q_h} \epsilon_N^F} \end{aligned}$$

In this equation, $\epsilon_N^{p_h}$ is the elasticity of price in entry, ϵ_M^d is the market elasticity of demand for hospital care in the private market, and ϵ_N^F is the elasticity of fixed costs with respect to number of firms in the market. The ratios $\frac{p_m - c_m}{p_m}$ and $\frac{p_h - c_h}{p_h}$ are price-cost margins in the public and private markets, respectively. The measure of cost-shifting is positive as

long as price is falling in N and $\frac{p_h - c_h}{p_h}$ is less than $|\epsilon_M^d|$.⁸ We do not attach a number to this long-run pass-through; the empirical section instead documents the closure-to-rival-price response directly.

2.4.2 Averted Exit: The Distressed Hospital's Own Price

This is the bargaining model behind Proposition 1, now turned to the averted-exit branch. Health-policy discussions of cost-shifting feature hospitals using their public shortfalls as leverage in negotiations with insurers, the lower public reimbursement serving as a rationale for higher private rates. The benchmark showed that this leverage is empty for a secure hospital. It has force only when the hospital's participation is at stake, meaning the cut has made the hospital unviable so that failing to agree means closing. In that case, as we now show, cost-shifting falls out of the simplest model of such a negotiation.

Bilateral Bargaining Model Consider a monopoly hospital and a monopoly insurer bargaining over hospital price. To focus attention on the bargaining, let us assume a very simple structure of consumer demand in the private sector. There is a population of consumers with measure 1. When they become ill (with probability q_h), consumers inelastically demand 1 unit of hospital care. Because this demand is inelastic and consumers are identical, there is no moral hazard or adverse selection, and the insurer optimally provides first-dollar coverage for hospital care. Ex ante, consumers are assumed willing to pay $W > c_h q_h + F - (p_m - c_m) q_m$ for insurance.

The monopolist insurer optimally charges W for insurance. Assume no underwriting costs to the insurer. Now, consider the negotiation between the insurer and the hospital. Each party's disagreement point is profit of zero.⁹ When agreement is reached at a

⁸This latter is a reasonable assumption. For example, if the pricing equilibrium of the game features price-setting by the hospital, then this is equivalent to assuming that the residual elasticity of demand facing the marginal firm is no less in absolute value than is the market elasticity of demand, a condition satisfied by many oligopoly pricing models.

⁹Notice that this is a strong assumption. It assumes that consumers can't consume hospital care without insurance and that insurers can't write insurance without an agreement with the hospital. Finally, it assumes that the hospital is unprofitable in the absence of an agreement and that it ceases operations in this case.

hospital price of p_h , each party earns a profit of:

$$\begin{aligned}\Pi_I &= W - p_h q_h \\ \Pi_H &= (p_m - c_m)q_m + (p_h - c_h)q_h - F\end{aligned}$$

The total surplus available to the negotiation is the sum of these two profits:

$$TS = W - c_h q_h + (p_m - c_m)q_m - F$$

The insurer and the hospital are negotiating over the hospital's price, p_h . Assuming simple Nash bargaining, the parties agree to split the surplus in half and set the hospital's price to give the hospital half of the total surplus:

$$(p_m - c_m)q_m + (p_h - c_h)q_h - F = \frac{1}{2} (W - c_h q_h + (p_m - c_m)q_m - F)$$

$$p_h = \frac{1}{2}c_h + \frac{1}{2q_h} (W - (p_m - c_m)q_m + F)$$

$$\frac{\partial p_h}{\partial p_m} = -\frac{q_m}{2q_h}$$

$$\theta = -\frac{\partial p_h}{\partial p_m} \frac{p_m}{p_h} \frac{p_h q_h}{p_m q_m} = \frac{1}{2}$$

Thus, the Nash bargaining model of hospital price determination yields a predicted cost-shift of 50%. The above assumes that the hospital and insurer have equal bargaining power. We can easily generalize away from this, however. Suppose that the hospital's relative bargaining power is $0 \leq \alpha \leq 1$. Now, the parties negotiate a price to make the

hospital's fraction of the surplus equal to α :

$$\begin{aligned}
(p_m - c_m)q_m + (p_h - c_h)q_h - F &= \alpha (W - c_h q_h + (p_m - c_m)q_m - F) \\
p_h &= (1 - \alpha)c_h + \frac{1}{q_h} [\alpha W + (1 - \alpha)(F - (p_m - c_m)q_m)] \\
\frac{\partial p_h}{\partial p_m} &= -(1 - \alpha) \frac{q_m}{q_h} \\
\theta &= -\frac{\partial p_h}{\partial p_m} \frac{p_m}{p_h} \frac{p_h q_h}{p_m q_m} = (1 - \alpha)
\end{aligned} \tag{5}$$

Perhaps counter-intuitively, cost-shifting is a decreasing function of the bargaining power of the hospital. When the hospital has no bargaining power, the insurer receives all of the surplus, and the hospital makes zero profit. In this case, because the hospital is kept constantly on the edge of bankruptcy, the insurer must fully compensate the hospital for any change in public reimbursement.

Conversely, when the hospital has all the bargaining power, the natural intuition of the economist matches reality: A hospital with full bargaining power sets the price that maximizes their profits from that insurer, and costs from other payers are irrelevant. Thus, the degree of cost-shifting in this story is an inverted, but direct measure of the relative bargaining power of the insurer. So, empirical estimates of cost-shifting parameters become estimates of the bargaining strength of insurers.

Financial fragility and the disagreement payoff The derivation above assumes that a hospital that fails to reach an agreement earns zero: it shuts down. That assumption is load-bearing, and isolating it separates two forces that the reduced form otherwise conflates. Let $R \equiv (p_m - c_m)q_m$ be the hospital's public net revenue and $D_H(R)$ its disagreement payoff—what it earns if the negotiation breaks down. Nash bargaining awards the hospital its disagreement payoff plus share α of the gains from trade, $\Pi_H = D_H(R) + \alpha (W - c_h q_h + R - F - D_H(R))$, which yields

$$\frac{\partial p_h}{\partial p_m} = -\frac{q_m}{q_h} (1 - \alpha) \rho, \quad \rho \equiv 1 - D'_H(R), \quad \text{so} \quad \theta^B = (1 - \alpha) \rho. \tag{6}$$

The public payment reaches the negotiated price only through the part of it that is at stake in the bargain, $\rho = 1 - D'_H(R)$: the fraction of a marginal public dollar the hospital would forgo by walking away. When disagreement means exit the hospital keeps none of R , so $D'_H(R) = 0$, $\rho = 1$, and $\theta^B = 1 - \alpha$, the case above. When a secure hospital could reject the contract, keep operating, and retain its public business, R enters its agreement and disagreement payoffs identically and cancels: $D'_H(R) = 1$, $\rho = 0$, and there is no cost-shifting, whatever its bargaining power—this is Proposition 1. Two distinct primitives thus govern the bargaining channel: financial fragility ρ sets whether the agreement matters for the hospital's survival, and the bargaining weight $1 - \alpha$ sets who absorbs the shock once it does. The two comparative statics are signed: $\partial\theta^B/\partial\rho = 1 - \alpha \geq 0$, so pass-through rises with fragility, while $\partial\theta^B/\partial\alpha = -\rho \leq 0$, so among fragile hospitals a hospital's own-price pass-through falls in its bargaining power. The object ρ is the share of public revenue at stake in this particular bargain—continuation dependence—rather than distance to exit as such; our predicted-closure-risk measure is its empirical proxy, not ρ itself. Full pass-through requires both $\rho = 1$ and $\alpha = 0$; proximity to exit is necessary but not sufficient. A simple microfoundation makes ρ continuous: if disagreement ends in exit with probability h rather than deterministically, then $D'_H(R) = 1 - h$ and $\rho = h$, so fragility is the off-path probability that a breakdown forces the hospital to close, ranging over $[0, 1]$. Because h is that disagreement-exit probability rather than the realized closure rate, our predicted-closure-risk moderator is tightly linked to ρ without equaling it.

2.5 Summary and Empirical Implications

The analysis yields five implications. First, standard models predict no cost-shifting, even with bargaining and market power: with the set of hospitals fixed, a public-payment cut moves neither the surplus a hospital and insurer bargain over nor the bargaining power that divides it, so the negotiated commercial price is unchanged, whatever the distribution of market power (Proposition 1). Cost-shifting is not a general feature of profit-maximizing hospital pricing.

Second, cost-shifting requires the exit margin. It appears only when the shock threatens a hospital's participation, and then through the insurer's response to that threat: faced with a hospital a cut has made unviable, the insurer either keeps it open on richer terms or lets it go and re-contracts with those that remain. These two resolutions define the pa-

per's two channels. In the averted-exit channel, bargaining lets a distressed hospital raise its own price. A public dollar reaches the negotiated price only to the extent the hospital keeps it under agreement but loses it in disagreement, so the cut lifts the negotiated private price only for a hospital whose participation is fragile, at the rate $\theta^B = (1 - \alpha)\rho$: the insurer's surplus share $(1 - \alpha)$ scaled by the hospital's financial fragility ρ . The hospital stays open and recoups part of the shortfall in its own commercial rate. In the realized-exit channel, a cut that pushes a marginal hospital past viability closes it, and the resulting rise in concentration lets its surviving rivals raise private prices; the exiting hospital recovers nothing, and the increase accrues to the hospitals that remain.

The two channels are conditional and touch different hospitals. Both are triggered only when the shocked hospital is near the exit margin; a secure hospital passes nothing through its own price. But they move different prices. The averted-exit channel raises the distressed hospital's own negotiated rate, and where that bargaining channel is isolated its own-price pass-through recovers the insurer's surplus share $(1 - \alpha)$ only once the hospital's fragility ρ is known. The realized-exit channel instead raises the prices of surviving rivals, which may themselves be financially secure, and its size tracks the market share a closure removes rather than any hospital's bargaining power. The two are substitutes for a given hospital: a large enough negotiated concession averts exit; if it does not, the hospital exits and the rival channel takes over.

These implications sharpen the empirical question. The prediction is not that cost-shifting rises uniformly with market power. Away from the exit margin a hospital's own price does not respond to its own public cut at any level of market power. The prediction is that cost-shifting appears where a hospital is near exit. A secure hospital passes nothing through its own price, but when a fragile rival exits, even a secure survivor's price can rise. What cost-shifting requires, then, is that some hospital in the market be near exit. A test has to isolate the hospitals for which exit is a live threat, and studies drawing samples of different financial composition should be expected to reach different conclusions.

The rest of the paper takes these predictions to hospital-level commercial prices, in the order the results section lays out; the data and empirical strategy behind each step follow next.

3 Data

The analysis links four sources at the hospital level: Medicare cost reports for finances and payer mix, RAND commercial-price estimates for the outcome, the Health Resources and Services Administration’s 340B registry for the payment shock, and a hospital-closure tracker for the exit margin. We describe these and then the sample.

Medicare Cost Reports. Every Medicare-participating hospital files an annual cost report. From the Healthcare Cost Report Information System we build a 2011–2023 panel of short-term and critical-access hospitals with operating and total margins, uncompensated-care and disproportionate-share exposure, payer mix (Medicare, Medicaid, and other discharge shares), bed size, ownership, and a charge-to-cost markup that contains no Medicare price and so is safe to use around Medicare-linked shocks. The cost reports also anchor the closure tracker and the market shares used to measure concentration.

RAND Commercial Prices. The outcome is the hospital-level commercial price from the RAND Hospital Price Transparency study (rounds 3–5, covering claims years 2017, 2019, and 2021). RAND reprices a large volume of employer and insurer claims to a common standard and reports a standardized commercial price—the allowed amount for a fixed bundle of services—separately for inpatient and outpatient care. We use the standardized price rather than the commercial-to-Medicare ratio because the ratio places Medicare in the denominator and would mechanically respond to a Medicare-linked shock; the standardized price is a commercial-dollar figure that does not.¹⁰ RAND’s price transparency study is a voluntary claims sample with thin commercial coverage of small, rural, and critical-access hospitals, so the exit-marginal hospitals we observe in RAND are a selected subset: only 61% of top-tercile exit-marginal hospitals carry a non-missing RAND outpatient price, against a higher share of more secure hospitals. This bears on external validity rather than on the within-hospital interaction, which is identified off the hospitals that do appear; Appendix Table 5 reports the round-by-round counts.

¹⁰RAND’s standardized price is the sum of claim-level allowed amounts divided by a service-mix index; it is not repriced against 340B-adjusted average sales price, so the 340B shock enters only through negotiated commercial rates rather than through the price construction.

The Exit Margin. Exit-marginality is a hospital’s predicted probability of closure, estimated from pre-shock information. We assemble closures from the Centers for Medicare and Medicaid Services provider tracker (confirmed closures with termination dates through 2019) and cross-check them against hospitals that stop filing cost reports; the two sources agree on 100% of tracker closures and on closure timing within one year in 95% of cases. Tracker confirmation runs only through 2019, while the RAND price waves extend to 2021, so a closure in 2020–21 is dropped from the survivor panel but its market is not flagged as treated; it then enters the rival-price test as an untreated control, which again pushes the estimated rival-price effect toward zero. We fit a closure model on 2014–2017 closures using only pre-2014 financials (margin, uncompensated-care share, size, ownership) and use its fitted value as a fixed, pre-treatment measure of how close each hospital is to exit. Because the model is estimated on closures that precede the 2018 shock and on financials that precede those closures, the moderator cannot be an artifact of the treatment.

Sample. The main pass-through test runs on the hospitals with a non-missing RAND commercial price, a defined 340B status, and a computable pre-shock closure hazard—an analytic sample of 3,336 hospitals; the two-way fixed-effects estimator then absorbs single-round hospitals, leaving 7,202 hospital-round observations across 41 states. Appendix B traces these restrictions step by step in an attrition table (Table 5). Prices are logged; we winsorize at the first and ninety-ninth percentiles in robustness checks. Table 1 reports summary statistics for the main analysis variables across the data sets.

4 Shocks

The empirical work rests on a public-payment shock that moves public revenue for an identifiable set of hospitals while leaving each hospital’s own commercial price outside its control.

Table 1: Summary statistics for the analysis data sets

	Mean	SD	P10	Median	P90	N
<u>Panel A. Hospital characteristics and finances (cost-report panel, 2011–2023)</u>						
Total margin	0.044	0.119	-0.088	0.041	0.186	58,956
Operating margin	-0.050	0.188	-0.256	-0.027	0.143	58,956
Uncompensated-care share	0.056	0.046	0.014	0.042	0.117	58,154
Medicaid discharge share	0.101	0.098	0.013	0.066	0.243	54,122
Medicare discharge share	0.389	0.169	0.190	0.367	0.631	58,761
Charge-to-cost markup	3.76	2.31	1.42	3.26	6.64	58,955
Staffed beds	143	169	19	74	364	58,956
For-profit	0.19	0.40	0.00	0.00	1.00	58,956
Critical-access hospital	0.29	0.45	0.00	0.00	1.00	58,956
<u>Panel B. Commercial outpatient prices (RAND standardized, Medicare-free; round sample, claims 2016–2022)</u>						
Standardized outpatient price (\$)	310	126	167	291	473	9,834
Log standardized outpatient price	5.658	0.400	5.120	5.673	6.159	9,834
<u>Panel C. Residual commercial inpatient prices (Wu 2009 method, 2013–2023)</u>						
Private inpatient price (\$)	21,866	21,367	6,366	14,558	45,824	44,946
Log private inpatient price	9.669	0.770	8.759	9.586	10.733	44,946
Private price-to-cost ratio	1.340	0.418	0.891	1.273	1.863	44,946
<u>Panel D. Shock exposure and market concentration</u>						
Exit-marginality hazard	0.027	0.031	0.007	0.016	0.057	56,838
340B-treated (DSH/RRC/SCH)	0.264	0.441	0.000	0.000	1.000	58,956
DSH only	0.231	0.422	0.000	0.000	1.000	58,956
Commercial insurer HHI (2013)	3,712	1,725	2,061	3,130	5,867	58,955
Hospital HHI (CBSA)	1,865	2,209	374	820	5,043	58,956

Note: Panels A, C, and D are hospital-year observations; Panel B is hospital-round. Panels A and D moments are computed on the Medicare cost-report panel (2011–2023); Panel B on the RAND standardized (Medicare-free) outpatient negotiated-price panel, the headline 340B outcome (round sample, claims 2016–2022); Panel C on the residual commercial inpatient price panel constructed following Wu (2009) (2013–2023). Continuous variables are winsorized at the 1st and 99th percentiles for the displayed moments; indicators are reported as sample means. Margins and shares are fractions of the relevant base; the markup is the charge-to-cost ratio; the exit-marginality hazard is the predicted pre-shock closure probability; the 340B-treated indicators are pre-2018 HRSA registry membership (DSH/RRC/SCH roster and the DSH-only subset), with registry non-matches coded to control; the commercial insurer HHI is the 2013 MLR commercial state-level value; the insurer and hospital HHIs are on the 0–10,000 scale.

4.1 The 340B Payment Shock

The 340B program lets qualifying hospitals purchase outpatient drugs at a discount. A 2018 Medicare rule cut the outpatient reimbursement for 340B-acquired drugs from average sales price plus 6 percent to average sales price minus 22.5 percent, a large reduction in the effective public payment for the outpatient drug care of exposed hospitals. We define treatment from the 340B registry: a hospital is exposed if it was a registered disproportionate-share, rural-referral, or sole-community provider in 2017, before the cut. On this definition about 24% of sample hospitals are exposed and 45% carry any 340B registration (Table 1, Panel D), in line with the roughly 40 to 45 percent of U.S. general hospitals that participate nationally. Because 96% of exposed registry hospitals are observed in RAND, the registry non-matches that fall to the control group are few, so any resulting misclassification attenuates the estimated interaction rather than threatening identification. The shock is outpatient-specific, which makes inpatient prices a natural outcome-specificity check.

5 Empirical Strategy

5.1 Neutrality Away from the Exit Margin

Before the exit-margin tests we check the benchmark that cost-shifting does not scale with market power. On the same Medicare-free RAND outpatient price and 2018 340B panel as the core test, we estimate

$$\log p_{it} = \beta (340B_i \times \text{Post}_t \times M_i) + \gamma (340B_i \times \text{Post}_t) + \delta (\text{Post}_t \times M_i) + \lambda_i + \tau_t + \varepsilon_{it}, \quad (7)$$

where M_i is, in turn, the standardized 2013 commercial insurer HHI (state), the predetermined hospital HHI, and their difference (bilateral relative power), λ_i are hospital fixed effects, and τ_t are round fixed effects. Proposition 1 predicts $\beta = 0$ for every M_i : away from the exit margin, no amount of market power turns a public cut into higher commercial prices. Each M_i proxies for the single surplus-division parameter α : insurer concentration, hospital concentration, and their difference are alternative readings of who holds bargaining power, and neutrality predicts $\beta = 0$ for all three because α governs only how

the surplus is split, not whether p_m enters it. Inference follows the same families as the core test, and we read a tightly estimated β near zero as the benchmark holding rather than as a failure to detect.

5.2 Pass-Through at Hospitals Near Exit

The core specification asks whether the 2018 340B outpatient shock raised commercial prices more at exit-marginal hospitals. For hospital i in RAND round t we estimate

$$\log p_{it} = \beta (340B_i \times \text{Post}_t \times H_i) + \gamma (340B_i \times \text{Post}_t) + \delta (\text{Post}_t \times H_i) + \lambda_i + \tau_t + \varepsilon_{it}, \quad (8)$$

on outpatient prices, where H_i is the standardized pre-shock closure hazard, Post_t indicates rounds after the 2018 cut, λ_i are hospital fixed effects, and τ_t are round fixed effects. The coefficient of interest is β : the extra pass-through of the payment cut at a hospital one standard deviation closer to exit. The theory predicts $\beta > 0$ and, at going-concern hospitals (low H_i), a pass-through near zero or negative. We commit to these signs before estimation. Because the shock is outpatient-specific, we estimate the same equation on inpatient prices as an outcome-specificity diagnostic (a response there would be a clean cross-service movement, not a mechanical artifact of the shock).

Identification rests on a triple-difference parallel-paths assumption: absent the shock, the gap in commercial-price growth between exit-marginal and secure hospitals would have evolved the same way among 340B hospitals as among non-340B hospitals. The chief threat is that the 2018 cut coincided with other pressures on safety-net hospitals, such as disproportionate-share (DSH) or uncompensated-care shocks, that also correlate with closure risk. Two features of the design speak to this. First, the outpatient-drug shock leaves inpatient prices unmoved (-0.0140 log points, $p = 0.589$), whereas a general safety-net squeeze would move both, so this outcome-specificity is the load-bearing rebuttal. Second, pass-through is near zero or negative at secure hospitals, so the interaction is not tracking a broad distress trend. We do not claim the non-340B group nets out DSH exposure, since DSH hospitals sit inside the treatment group; we lean on the inpatient null instead. With one pre-shock round we cannot test pre-trends inside the RAND panel, so we report the inpatient diagnostic and, in the appendix, an annual check on the charge-to-cost markup, and rest on the design’s falsifications rather than on an untestable lead.

We report four inference families for β : hospital-clustered and state-clustered standard errors with a $t(G - 1)$ reference, a restricted-null wild-cluster bootstrap at the state level, and randomization inference that permutes the closure-hazard labels across hospitals. We treat the state cluster as our preferred standard—the shock and the concentration moderator both vary at or above the state level—and report the others so that no single p -value stands alone.

5.3 Realized Exit: Closures and Surviving Rivals' Prices

The realized-exit channel predicts that when a hospital closes, its surviving rivals raise commercial prices. We match each surviving RAND hospital to the confirmed closures in its market and estimate

$$\log p_{it} = \beta \text{Closure}_{mt} + \lambda_i + \tau_t + \varepsilon_{it}, \quad (9)$$

where Closure_{mt} is either an indicator that at least one hospital has closed in market m by year t (extensive margin) or the cumulative discharge share those closures remove (dose), λ_i are hospital fixed effects, and τ_t are year fixed effects. Markets are defined at the metropolitan (CBSA) and county levels, and ever-exiting hospitals are dropped so the outcome is a survivor's price. The theory predicts $\beta > 0$, and we report both the extensive-margin indicator and the cumulative-share (dose) coefficient. Because closures cluster geographically, we report CBSA-clustered and state-clustered standard errors, a state-level wild-cluster bootstrap, and randomization inference that reassigns the market treatment profile across markets; and we verify that soon-to-close markets are not already priced above never-close markets before the closure.

6 Results

Our central prediction has two parts. Away from the exit margin, cost-shifting is absent no matter how market power is distributed. It appears only where a hospital is near exit, and there it shows up in different prices depending on whether the exit is averted or realized. We check the neutrality benchmark first. We then test the averted-exit branch, in

which a distressed hospital raises its own negotiated price, using the 2018 340B outpatient cut, and the realized-exit branch, in which surviving rivals raise prices after a competitor closes, using hospital closures.

6.1 Neutrality Away from the Exit Margin

The benchmark is that a public-payment cut leaves commercial prices unmoved when no hospital's participation is at stake, at any level of insurer concentration (Proposition 1). Table 2 tests it on the same Medicare-free RAND outpatient price and the same 2018 340B treatment as the core exit-marginality result, replacing the exit-marginality moderator with three measures of market power (the neutrality moderators are defined for more hospitals than the exit-marginality moderator, which additionally needs an estimable pre-2014 closure hazard, so the analytic sample is somewhat larger). The pooled 340B effect is null, and none of the market-power interactions is distinguishable from zero: the extra pass-through of the cut is 0.0088 log points per standard deviation of commercial insurer concentration, -0.0015 for hospital concentration, and 0.0021 for bilateral relative power. These are precise nulls. The 95 percent confidence interval for the insurer-power interaction reaches only 0.030 log points, about the size of the exit-marginality effect itself (0.0261), so the data rule out an insurer-power channel materially larger than the exit channel rather than merely failing to detect one. Insurer HHI varies only at the state level, but the hospital-concentration and relative-power moderators vary at the finer market level and are also precise nulls, so the flat response is not an artifact of measurement coarseness. Cost-shifting does not scale with market power, as neutrality predicts, and the same interaction is absent for inpatient prices. Pass-through does appear, but only where hospitals are near exit, as the rest of the section shows.

6.2 Pass-Through Rises with Proximity to Exit

Table 3 reports the core test. The triple interaction is positive and, at the state-cluster level, significant: a one-standard-deviation increase in predicted closure risk makes the commercial-price response to the 340B outpatient cut 0.0261 log points more positive. Splitting by round, the effect is concentrated in the first post-shock round (0.0289 log points). The comparison with the going concerns is the theory's other half. At hospitals with lower predicted closure risk the pass-through is -0.0271 log points, which is no

Table 2: Fixed-participation neutrality: the 2018 340B payment cut does not pass through more strongly where market power is greater

	Coefficient	Std. error	State p	Bootstrap p
<u>Pooled 340B effect on the RAND outpatient price</u>				
Treat $\times$ Post	−0.0188	(0.0121)	0.125	—
<u>Interaction with market power: Treat $\times$ Post $\times$ moderator</u>				
$\times$ Insurer HHI	0.0088	(0.0110)	0.425	0.414
$\times$ Hospital HHI	−0.0015	(0.0077)	0.851	0.837
$\times$ Relative power (insurer − hospital)	0.0021	(0.0093)	0.822	0.837
<u>Inpatient specificity: Treat $\times$ Post $\times$ Insurer HHI = −0.0003, p = 0.987</u>				
Observations	8,913 (outpatient hospital-rounds)			
Hospitals	4,127			
State clusters	50			
Hospital FE, Round FE	Yes			

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; $t(G-1)$ reference with G state clusters; bootstrap p is the state Rademacher wild-cluster bootstrap (999 draws). The outcome is the log RAND standardized (Medicare-free) outpatient negotiated price, with the same 340B treatment as Table 3. Insurer HHI is the 2013 commercial insurer HHI (state); hospital HHI is the predetermined HCRIS hospital HHI; relative power is their difference; each is standardized. Standard errors clustered on state in parentheses.

positive cost-shifting at all, and that is what the theory predicts for a hospital whose participation constraint does not bind. Pooling across all 340B hospitals, exit-marginal and secure together, the effect is null. That pooled null is the fixed-participation benchmark of Proposition 1, recovered here as an average over hospitals on both sides of the exit margin.

Three diagnostics support the interpretation. Inpatient prices, which the outpatient drug shock does not touch, show no interaction (−0.0140 log points), consistent with a response specific to the shocked outpatient service rather than a market-wide movement. The result survives permuting which hospitals carry the exit-marginality label, so it attaches to the specific financially marginal hospitals rather than to the moderator’s distribution. And it holds when the vertically integrated hospital systems that drive unrelated pricing anomalies are excluded.

Table 3: Exit marginality and the pass-through of the 2018 340B payment cut into RAND outpatient negotiated prices

	(1) Pooled triple	(2) Round dynamics	(3) Lower- risk
Treat $\times$ Post $\times$ Exit marginality	0.0261** (0.0107)		
$\times$ 2019 (round 4)		0.0289* (0.0143)	
$\times$ 2021 (round 5)		0.0183 (0.0113)	
Treat $\times$ Post (subgroup)			−0.0271* (0.0142)
<u>Inference on column (1) triple interaction:</u>			
Hospital (CCN) clusters, $t(G-1)$	$p = 0.084$		
State clusters, $t(G-1)$	$p = 0.020$		
State wild-cluster bootstrap	$p = 0.037$		
Hazard-permutation RI	$p = 0.022$		
<u>Inpatient outcome specificity:</u> Treat $\times$ Post $\times$ Exit marginality = −0.0140, $p = 0.589$			
Observations	7,202	7,202	6,447
Hospitals	2,590	2,590	2,300
State clusters	41	41	41
Hospital FE	Yes	Yes	Yes
Round FE	Yes	Yes	Yes

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; $t(G-1)$ reference with G state clusters. The outcome is the log RAND standardized (Medicare-free) outpatient negotiated price. Exit marginality is a hospital's predicted closure hazard, estimated from pre-2014 financials and 2014–2017 closures, standardized (columns 1–2) or split at its top tercile (column 3). Post = 1 for RAND rounds 4–5 (2019, 2021); the 2018 340B outpatient-drug payment cut is the shock. Column (1) is the pooled triple interaction; column (2) traces it round by round (2017 = reference); column (3) is the Treat $\times$ Post effect within the lower-risk hospitals, those below the top tercile of exit marginality. The Hospitals row counts the hospitals observed in ≥ 2 RAND rounds that identify the two-way fixed-effects estimates; the analytic sample before single-round hospitals are absorbed is 3,336 hospitals (Appendix Table 5). Standard errors clustered on state in parentheses.

6.3 Realized Exit: Closures Raise Surviving Rivals' Prices

The realized-exit channel makes a different prediction about a different set of hospitals. When a hospital closes it recovers nothing, but its surviving rivals raise prices as competition weakens. Matching each RAND rival to the closures in its market, we find that

after a hospital closes its surviving rivals raise commercial outpatient prices by 3.9% at the metropolitan market definition and 5.3% at the finer county definition. Soon-to-close markets are not already priced above never-close markets, so the increase follows the closure rather than preceding it. The slope of rival prices in the cumulative closed market share is 0.595, so a larger displaced share raises survivors' prices more. These increases are economically substantial, but their statistical precision varies across the market definition (Table 4).

Table 4: Hospital closure and surviving rivals' outpatient price

	(1) CBSA	(2) County
	Market definition	
<u>Panel A. Any hospital closure (extensive margin)</u>		
Any closure	+0.0389** (0.0185)	+0.0529** (0.0215)
<u>Panel B. Cumulative closed discharge share</u>		
Cumulative closed share	+0.5947** (0.2872)	+0.0538 (0.0403)
Observations	7,119	7,097
Markets (clusters)	429	1,271
Hospital FE	Yes	Yes
Year FE	Yes	Yes

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; $t(G - 1)$ reference with G clusters. Outcome is log RAND outpatient standardized price at surviving rival hospitals (ever-exiting hospitals dropped), 2017/2019/2021 claims waves. Treatment is a tracker-confirmed closure with a computable concentration dose in the same market, 2013–2021. The CBSA column uses the native discharge-share dose; the county column recomputes the closed-hospital size share from HCRIS discharges at that geography, so Panel B magnitudes are not on the CBSA scale.

7 Conclusion

Standard theory rejects cost-shifting for a comfortable profit-maximizing hospital, and it is right to. Hold the set of hospitals fixed and a public-payment cut moves neither the surplus a hospital and insurer bargain over nor the power that divides it, so a hospital

already charging its optimal private price has nothing to shift, whatever its bargaining power. Cost-shifting requires the exit margin. When a cut threatens a hospital's survival the insurer has to choose. It can keep the hospital open on richer terms, in which case the distressed hospital's own rate rises by the insurer's share of the surplus scaled by the hospital's fragility, or it can let the hospital close, in which case the surviving rivals' rates rise as capacity leaves the market. Cost-shifting is therefore neither a myth nor a constant. It is a feature of how close a hospital sits to going out of business.

That cost-shifting turns on how close a hospital is to exit could settle a literature that never has. Pass-through appears where hospitals are near exit and vanishes where they are secure, so studies drawing samples of different financial composition reach different conclusions without contradicting one another. Reading cost-shifting this way also scopes what any given estimate identifies. An own-price pass-through in the averted-exit branch recovers the insurer's share of the surplus $(1 - \alpha)$ only once the hospital's fragility is known, while a rival-price response to a closure measures something else entirely, the competitive spillover from lost capacity. The policy implication follows whichever branch is doing the work. Where distress-driven exit is the mechanism, keeping hospitals viable is what limits commercial-price spillovers. Where surviving rivals' prices are, the spillover depends on the concentration a closure creates, and the tools that act on concentration are the relevant ones.

Two limitations bound these claims. Identification comes from specific shocks: the 2018 340B outpatient cut and a set of confirmed hospital closures. The estimated magnitudes are therefore local to those settings, so extrapolating the 340B outpatient slope to broad rate cuts would be an extrapolation rather than a forecast; what generalizes is the qualitative prediction that pass-through concentrates at the exit margin and is absent for secure hospitals. Because the commercial-price data are a selected sample of hospitals, the estimates also speak to the hospitals observed in them rather than to the universe of providers.

The framework yields a sharp and testable prediction: a hospital's own price responds to its own public cut only when it is near exit, and not at all when it is a secure going concern, while a closure raises the prices of surviving rivals, which may themselves be secure. Pairing hospital-level commercial prices with public-payment shocks and a measure of exit risk, the evidence above is consistent with this predicted incidence: away from

the exit margin the price response does not rise with insurer market power, the own-price response concentrates at the exit-marginal hospitals, and surviving rivals' prices rise after a competitor closes.

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A Proof of Fixed-Participation Neutrality

This appendix proves the neutrality result stated in Section 2.2.

Proof of Proposition 1. Let $R \equiv (p_m - c_m)q_m$ be the hospital's public net revenue, with $dR/dp_m = q_m$, and $D_H(R)$ its disagreement payoff. Total surplus is $TS = W - c_h q_h + R - F$, and the gains from trade are $G(R) = W - c_h q_h + R - F - D_H(R)$, with $G'(R) = 1 - D'_H(R)$. The asymmetric Nash bargain with hospital weight α awards the hospital its disagreement payoff plus share α of the gains from trade:

$$R + (p_h - c_h)q_h - F = D_H(R) + \alpha G(R). \quad (10)$$

Differentiating (10) in p_m (holding $\alpha, c_h, q_h, q_m, F, W$ fixed) gives $q_m + (\partial p_h / \partial p_m)q_h = D'_H(R)q_m + \alpha(1 - D'_H(R))q_m$, so

$$\frac{\partial p_h}{\partial p_m} = -\frac{q_m}{q_h}(1 - \alpha)(1 - D'_H(R)) = -\frac{q_m}{q_h}(1 - \alpha)\rho, \quad \rho \equiv 1 - D'_H(R). \quad (11)$$

A hospital that rejects the contract but continues to operate retains its public revenue R in the disagreement state, so $D'_H(R) = 1$ and $\rho = 0$; then $\partial p_h / \partial p_m = 0$ and $\theta = 0$ for every $\alpha \in [0, 1]$. Its public revenue raises the agreement and disagreement payoffs by the same amount, dropping out of the gains from trade G ; and α is a primitive of market structure, not of the payment rate. Neither the divided surplus nor the bargaining weight responds to p_m , so neither does the negotiated price. $\square$

The exit margin overturns the result by breaking $D'_H(R) = 1$. If disagreement means closure, the hospital forfeits its public revenue, so $D'_H(R) = 0$, $\rho = 1$, and $\theta^B = 1 - \alpha > 0$: the averted-exit branch of Section 2.4.2. Intermediate continuation dependence, $0 < \rho < 1$, interpolates between the two.

B Sample Construction

Table 5 traces the sample restrictions that define the core pass-through test, from the full set of RAND outpatient hospital-round records down to the two-way fixed-effects estimation sample. Each row imposes the listed restriction on top of those above it.

Table 5: Sample construction for the core pass-through test

	Hospitals	Hospital-round obs.
RAND HPT outpatient hospital-round records (rounds 3–5)	4,439	11,348
with a non-missing, positive standardized commercial price	4,127	9,834
linked to a Medicare cost-report hospital with a pre-2018 baseline	3,360	7,998
with a computable pre-shock closure hazard (analytic sample)	3,336	7,948
observed in ≥ 2 RAND rounds (identifies the hospital $\times$ round FE)	2,590	7,202

Note: Restrictions defining the estimation sample for the core 340B $\times$ exit-marginality test on RAND standardized outpatient commercial prices (Table 3). Each row applies the listed restriction on top of those above it. Every hospital in the RAND panel has a defined 340B registry status, so treatment assignment removes no observations; the binding restrictions are a valid commercial price and a computable pre-2018 closure hazard, the latter requiring a linked Medicare cost report with non-missing pre-2014 financials. The fourth row is the analytic sample (3,336 hospitals); the two-way hospital $\times$ round fixed-effects estimator then absorbs hospitals observed in a single RAND round, leaving 7,202 hospital-round observations across 41 state clusters.

C Robustness of the Exit-Marginality Result

Table 6 reports the exit-marginality triple interaction under alternative moderators (Panel A) and across alternative samples, treatment definitions, and outcome transformations (Panel B). The estimate is stable in sign and magnitude, and the uncompensated-care-share placebo moderator is null.

Table 6: Moderator transparency and robustness of the exit-marginality triple interaction (RAND outpatient price)

	Coefficient	Std. error	State p	Bootstrap p
<u>Panel A. Alternative exit-marginality moderators</u>				
Primary closure hazard (full logit)	0.0261**	(0.0107)	0.020	0.037
Margin + uncompensated-care hazard	0.0225**	(0.0097)	0.026	0.027
Financials-only hazard	0.0376*	(0.0197)	0.064	0.098
Closure hazard, leave-one-state-out cross-fit	0.0271**	(0.0107)	0.015	0.031
Operating margin, bottom tercile	0.0402*	(0.0236)	0.096	0.099
Uncompensated-care share, top tercile	0.0018	(0.0237)	0.939	0.956
<u>Panel B. Robustness (primary hazard moderator)</u>				
Drop round 5 (COVID, 2021)	0.0278*	(0.0146)	0.064	0.075
Drop critical-access hospitals	0.0264**	(0.0100)	0.012	0.026
340B DSH-only treatment	0.0173	(0.0114)	0.139	0.160
Hazard-decile dose	0.0021	(0.0036)	0.562	0.602
Exclude vertically integrated states (UT/MN/ND)	0.0254**	(0.0106)	0.022	0.040
Winsorize price 1/99	0.0202*	(0.0101)	0.051	0.073

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; $t(G-1)$ reference with G state clusters. Each row reports the triple interaction $\text{Treat} \times \text{Post} \times \text{moderator}$ from a separate hospital- and round-fixed-effects regression on the log RAND standardized outpatient price. Panel A varies the exit-marginality moderator; the uncompensated-care-share moderator (last row) is the placebo moderator. Panel B holds the primary closure-hazard moderator fixed and varies the sample, treatment definition, moderator scaling, and outcome winsorization. Bootstrap p is the state wild-cluster bootstrap. Standard errors clustered on state.

D The Charge-to-Cost Markup Around the 2018 Cut

Figure 1 plots two annual diagnostics for the 340B test. Panel A shows the triple interaction by RAND round; Panel B shows the high- versus low-hazard difference in the charge-to-cost markup, which embeds no Medicare price, by year. The markup difference is flat and near zero before the 2018 cut, the annual check referenced in the empirical strategy.

Figure 1: Round dynamics and the annual charge-to-cost markup for exit-marginal versus secure 340B hospitals. Panel A: the $\text{Treat} \times \text{exit-marginality}$ interaction by RAND round. Panel B: the high- versus low-hazard difference in the log charge-to-cost markup by year (reference 2017), flat and near zero before the 2018 cut.

